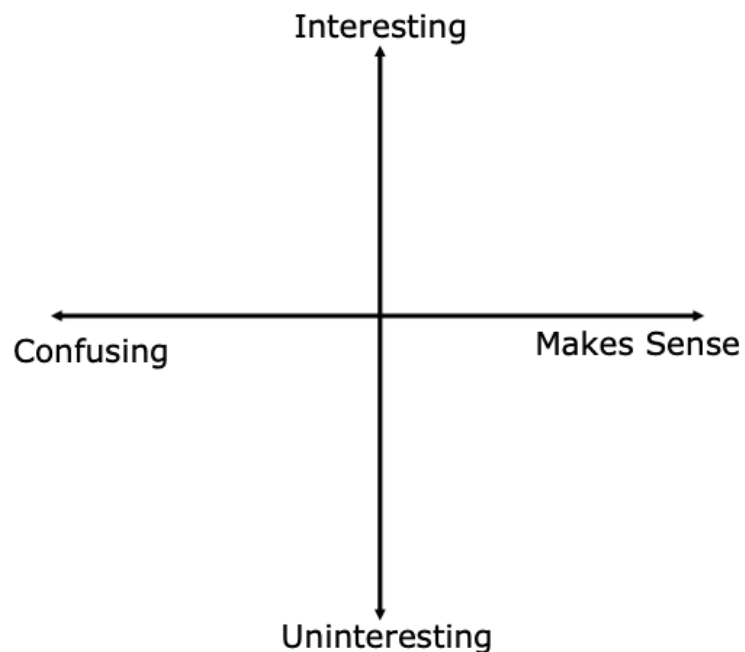




1.3.6 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate where you feel you are on today's learning target: "I can evaluate a function for a specific input value of the domain and interpret it in a real-world context."



Unit 1, Lesson 4: Applying the Laws of Exponents



Benchmark

MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.



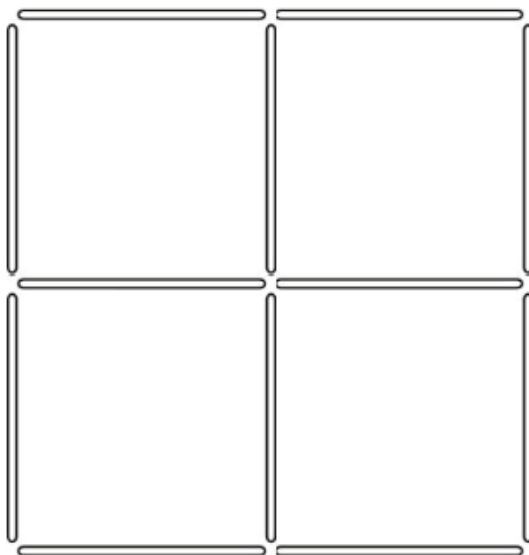
Learning Targets

- ☐ I can apply the Laws of Exponents to write equivalent numerical expressions.
- ☐ I can evaluate square roots and cube roots.



1.4.1 Warm-Up: Toothpick Squares

1. Make two squares of different sizes by removing two toothpicks.



1.4.2 Refresher: Laws of Exponents

1. Using the given exponent law, complete the table by writing an equivalent exponential expression.

Exponent Law	Expression	Equivalent Expression
Product of Powers $a^m \cdot a^n = a^{m+n}$	$6^5 \cdot 6^2$	
Quotient of Powers $\frac{a^m}{a^n} = a^{m-n}$	$\frac{5^{14}}{5^2}$	
Power of a Power $(a^m)^n = a^{m \cdot n}$	$(4^{10})^3$	
Power of a Product $(ab)^m = a^m \cdot b^m$	$(5 \cdot 2)^4$	
Power of a Quotient $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$	$\left(\frac{3}{11}\right)^2$	
Negative Exponents $a^{-1} = \frac{1}{a}$ $\left(\frac{a}{b}\right)^{-1} = \frac{b}{a}$ $\left(\frac{a}{b}\right)^{-m} = \frac{b^m}{a^m}$	9^{-1} $\left(\frac{5}{7}\right)^{-1}$ $\left(\frac{13}{4}\right)^{-3}$	
Identity Exponent $a^1 = a$	572^1	
Zero Exponent $a^0 = 1$	572^0	



1.4.3 Guided Instruction: Building on Square Roots and Cube Roots

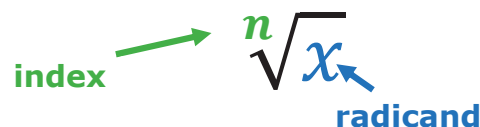
To square a number means to

- multiply it by itself.
- multiply it by 2.

$$7^2 = \underline{\quad} \cdot \underline{\quad} = \underline{\quad}$$

$$(-7)^2 = \underline{\quad} \cdot \underline{\quad} = \underline{\quad}$$

The inverse of squaring a number is taking the square root ($\sqrt{\quad}$).



To evaluate $\sqrt[2]{25}$, also written as $\sqrt{25}$, ask, "What times itself equals 25?"

$$5 \cdot 5 = 25 \text{ or } -5 \cdot -5 = 25$$

Therefore, $\sqrt{25}$ is 5 and -5 .



All radicands greater than or equal to 0, $x \geq 0$, have two square roots, a positive and a negative. The positive square root is referred to as the **principal square root**.

The square root of 16, written as $\sqrt{16}$, is 4 and -4 because both 4^2 and $(-4)^2$ are equal to 16.

To cube a number means to

- multiply the number by 3.
- triple the number.
- multiply the number by itself three times.

$$6^3 =$$

$$(-6)^3 =$$

The cube root of a number is a value that when multiplied by itself three times gives that number.

$$27 = \underline{\quad}^3 \text{ so } \sqrt[3]{27} = \underline{\quad}$$

$$-27 = \underline{\quad}^3 \text{ so } \sqrt[3]{-27} = \underline{\quad}$$

The cube root of 64, written as $\sqrt[3]{64}$, is 4 because 4^3 is 64.

$$4 \cdot 4 \cdot 4 = 64$$



1.4.4 Your Turn!

1. Find the value(s) of each.

a. $\sqrt{121} = \underline{\hspace{2cm}}$

b. $\sqrt[3]{125} = \underline{\hspace{2cm}}$

2. Work with a partner to complete the chart.

Expression	Exponent Law	Equivalent Expression
$8^2 \cdot 8^4$	Product of Powers	
	Product of Powers	10^7
$(7^5)^3$	Power of a Power	
	Power of a Power	4^{32}
$\frac{2^{12}}{2^{10}}$	Quotient of Powers	
	Quotient of Powers	6^9
$(3 \cdot 7)^5$	Power of a Product	
	Power of a Product	$13^5 \cdot 2^5$

$\left(\frac{1}{7}\right)^4$	Power of a Quotient	
	Power of a Quotient	$\frac{6^3}{25^3}$
$\left(\frac{2}{5}\right)^{-2}$	Negative Exponents	
$11^3 \cdot 11^{-5}$	Product of Powers	
	Negative Exponents	$\frac{1}{45}$

3. Marques, Pablo, and Tandice used the product of powers law to write an equivalent expression for $5^{-7} \cdot 5^6$. Each student's equivalent expression is shown in the table.

Marques	Pablo	Tandice
$\frac{1}{5}$	-5	5^{-1}

- Which student's answer is not an equivalent expression for $5^{-7} \cdot 5^6$?
- Which student also applied the negative exponent law?

4. Find the value of the variable(s) that makes the equation true.

a. $7^{10} = 7^5 \cdot 7^f$

b. $\sqrt[3]{h} \cdot \sqrt[3]{h} \cdot \sqrt[3]{h} = -13$

$f = \underline{\hspace{2cm}}$

$h = \underline{\hspace{2cm}}$

c. $\sqrt{m} \cdot \sqrt{m} = 26$

d. $\frac{7^5}{6^5} = \left(\frac{6}{7}\right)^y$

$m = \underline{\hspace{2cm}}$

$y = \underline{\hspace{2cm}}$

e. $35^{12} = (35^x)^2$

f. $\left(\frac{9}{10}\right)^{-1} = \frac{c}{r}$

$x = \underline{\hspace{2cm}}$

$c = \underline{\hspace{2cm}}$

$r = \underline{\hspace{2cm}}$

g. $18^0 = \frac{18^w}{18^7}$

$w = \underline{\hspace{2cm}}$



1.4.5 Wrap-Up: Cool Down

1. Use the symbols $<$, $>$, or $=$ to compare each pair of values. Then, explain your choice.

Expression	Explanation
$2^3 \underline{\hspace{2cm}} 2 \cdot 3$	
$5^2 \underline{\hspace{2cm}} 2^5$	
$100^1 \underline{\hspace{2cm}} 1^{100}$	